

Project Testing

Functional & Performance Testing

Date	11 July 2026
Team ID	SWTID-2026-9508
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

The tables below record the actual functional and performance test results executed against the Credit Card Approval Prediction System's Flask application, using its test client to simulate real HTTP requests end-to-end.

Functional Test Results:

Test Case ID	Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
FT-01	Application form loads correctly	Send GET request to '/'	Form page returns HTTP 200 with all input fields	HTTP 200 returned; form rendered with all fields	Pass
FT-02	Valid categorical dropdown values	Load form and inspect dropdown options for Gender, Married, Employed, etc.	All dropdowns populated with valid encoded categories	All category options rendered correctly from CATEGORY_OPTIONS	Pass
FT-03	Missing required field validation	Submit form to '/predict' with the Income field omitted	Form redisplayes with a clear 'Missing required field' error	Error message displayed: "Missing required field(s): Income"	Pass
FT-04	Invalid categorical value handling	Submit form with Gender set to an out-of-range value ('zzz')	Request is rejected with a validation error instead of crashing	ValueError caught; form redisplayed with error message	Pass
FT-05	Successful prediction (Approved case)	Submit form with a strong applicant profile (high income, no prior default, employed)	Result page shows 'Approved' with a confidence score	Result: Approved, Confidence: 98.99%	Pass
FT-06	Successful prediction (Rejected case)	Submit form with a weaker applicant profile	Result page shows 'Rejected' with a confidence score	Result: Rejected, Confidence: 1.33%	Pass

Test Case ID	Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
FT-07	Custom 404 error page	Request a non-existent route ('/this-route-does-not-exist')	Custom on-brand 404 page returned instead of Flask's default error	HTTP 404 returned with custom 404.html template	Pass

Performance Test Results:

Test Case ID	Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
PT-01	Single prediction response time	Measure average response time across 20 sequential POST /predict requests	Should be under 3 seconds per prediction	Average response time: 5.4 milliseconds per request	Pass
PT-02	Repeated request stability	Send 50 rapid, back-to-back POST /predict requests and verify all succeed	All requests should return HTTP 200 without errors or slowdown	All 50/50 requests returned HTTP 200 in 0.22 seconds total	Pass
PT-03	Model load time at startup	Measure time to load model.pkl, encoders.pkl, and scaler.pkl when the Flask app starts	Model artifacts should load once at startup with no noticeable delay	All artifacts loaded successfully at startup, under 1 second	Pass

Summary:

7 of 7 functional test cases passed and **3 of 3 performance test cases passed**. The application correctly handles valid and invalid input, returns accurate predictions with confidence scores, serves a custom 404 page for invalid routes, and responds well within acceptable performance thresholds (average 5.4ms per prediction, far under the 3-second target).